

This artifact is a text-based Hangman game written in C++ that I built a few years ago as a final project for CIS 125 at Berkshire Community College. The player tries to guess a secret word one letter at a time with the word either typed in by another player or randomly pulled from a text file by the computer. I chose this project for my ePortfolio because it gave me a real opportunity to show growth in how I think about data structures not just whether the code runs. The most meaningful change was swapping out the vector that tracked guessed letters for an unordered set. This means the program no longer must scan through every previous guess one by one and it just checks instantly. I also replaced the raw character arrays with strings which handle memory automatically and make the whole thing cleaner to read and work with.

As far as course outcomes go, I think this enhancement hit what I was aiming for. The switch to an unordered set directly demonstrates understanding of how data structure choice affects efficiency, and moving away from raw arrays shows I can recognize when manual memory management is unnecessary and honestly a little risky. Going forward I'd like to work with larger data sets where these kinds of decisions have a more visible impact since with a small game like this the difference is more about correctness and cleanliness than raw speed. Working through this also reminded me that good code isn't just about getting the right answer. It's about making efficient decisions along the way. When I first wrote this, I just used what felt familiar which is how most beginners approach it. Going back and asking whether each piece was the right tool for the job was a different mindset and the biggest challenge was making sure I only changed what was needed to change without breaking anything else. That process of tracing how everything connected ended up being just as valuable as the improvements themselves.